COHOMOLOGY OF A COHERENT MODULE ON A PROPER ALGEBRAIC SPACE OVER A NOETHERIAN AFFINE BASE IS A FINITE MODULE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-spaces-cohomology/preamble.tex is excerpted verbatim as three spans, lines 8-12 (verbatim package and the project's own reference/slogan/history display environments), lines 14-18 (Xy-pic setup) and lines 154-217 (theorem declarations and mathematical macros). The wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- Coherence of the higher direct images of a coherent module under a proper morphism. Same chapter spaces-cohomology.tex, label lemma-proper-pushforward-coherent (invoked at proof line 4020). It is the affine case of this lemma that is being used: already knowing $R^if_*\mathcal{F}$ coherent gives the finiteness conclusion.
- Quasi-coherence of the higher direct images along a quasi-compact quasi-separated morphism. Same chapter spaces-cohomology.tex, label lemma-higher-direct-image (invoked at proof line 4021). It makes $R^if_*\mathcal{F}$ a quasi-coherent module on $\Spec(A)$, hence the module associated to its global sections.
- The Leray identification of cohomology with global sections of a higher direct image, and vanishing on an affine scheme. Cohomology on Sites, label sites-cohomology-lemma-apply-Leray (invoked at proof line 4026) and Cohomology of Schemes, label coherent-lemma-quasi-coherent-affine-cohomology-zero (invoked at proof line 4028). Together they give $H^i(X, \mathcal{F}) = \Gamma(\Spec(A), R^if_*\mathcal{F})$.
- Coherence on a Noetherian affine base is finite generation of the global sections. Same chapter spaces-cohomology.tex, label lemma-coherent-Noetherian (invoked at proof line 4030). It turns the coherence of $R^if_*\mathcal{F}$ into finite generation of $H^i(X, \mathcal{F})$ over A.

Date: stacks-project:spaces-cohomology.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 01XB: sources/stacks-chow/tags.txt, source lines 2512–2512. Navigation pointer for \ref{coherent-lemma-quasi-coherent-affine-cohomology-zero} used inside the counted span (line 4029); the argument that consumes it is retained in the span.

Source reference. Original lemma 07UB: sources/stacks-spaces-cohomology/pinned-original-source.tex, source lines 2094–2094. Navigation pointer for \ref{lemma-coherent-Noetherian} used inside the counted span (line 4030); the argument that consumes it is retained in the span.

Source reference. Original lemma 0720: sources/stacks-spaces-cohomology/pinned-original-source.tex, source lines 339–339. Navigation pointer for \ref{lemma-higher-direct-image} used inside the counted span (line 4021); the argument that consumes it is retained in the span.

Source reference. Original lemma 08AR: sources/stacks-spaces-cohomology/pinned-original-source.tex, source lines 3835–3835. Navigation pointer for \ref{lemma-proper-pushforward-coherent} used inside the counted span (line 4020); the argument that consumes it is retained in the span.

Source reference. Original lemma 0733: sources/stacks-chow/tags.txt, source lines 9047–9047. Navigation pointer for \ref{sites-cohomology-lemma-apply-Leray} used inside the counted span (line 4026); the argument that consumes it is retained in the span.

Lemma 0.1. *Let A be a Noetherian ring. Let $f : X \to \mathrm{Spec}(A)$ be a proper morphism of algebraic spaces. Let $\mathcal{F}$ be a coherent $\mathcal{O}_X$-module. Then $H^i(X, \mathcal{F})$ is finite A-module for all $i \geq 0$.*

Proof. This is just the affine case of Lemma 08AR. Namely, by Lemma 0720 we know that $R^i f_* \mathcal{F}$ is a quasi-coherent sheaf. Hence it is the quasi-coherent sheaf associated to the A-module $\Gamma(\mathrm{Spec}(A), R^i f_* \mathcal{F}) = H^i(X, \mathcal{F})$. The equality holds by Cohomology on Sites, Lemma 0733 and vanishing of higher cohomology groups of quasi-coherent modules on affine schemes (Cohomology of Schemes, Lemma 01XB). By Lemma 07UB we see $R^i f_* \mathcal{F}$ is a coherent sheaf if and only if $H^i(X, \mathcal{F})$ is an A-module of finite type. Hence Lemma 08AR gives us the conclusion. □